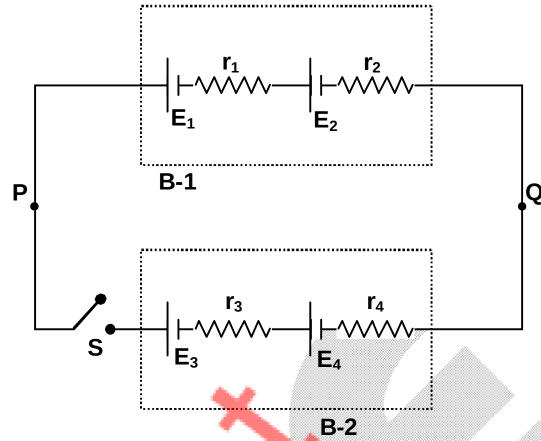
4. A battery B-1 is made by using two cells of emf E_1 and E_2 and internal resistance r_1 and r_2 respectively. Another battery B-2 is made by using different cells of emf E_3 and E_4 and internal resistance r_3 and r_4 respectively. Now they are connected with points P and Q as shown in the figure. All cells are rechargeable. Now switch is closed at t = 0, then choose the INCORRECT option.



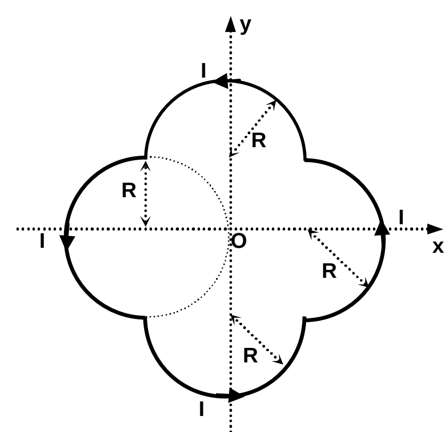
(A) The potential difference across both the battery will be equal.
(B) Rate of heat dissipation in both batteries is equal.
(C) One battery will charge the other battery.
(D) Rate of loss in chemical energy of one battery is more than the rate of gain in chemical energy of the other.

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A particle is projected at an angle of θ with the horizontal, with initial speed u. If the magnitude of velocity of projectile motion and time are related as $v^2 - 100t^2 + 400\,t - 800 = 0$. Then which of the following is correct. (take $g = 10\ m/s^2$)
(A) Angle of projection is 45°.
(B) Maximum height attained by particle is 20 meter.
(C) Range of the particle is 80 meter.
(D) Projection velocity is 40 m/sec.

6. A system of four semi-circular wires of radius a , carrying current I as shown in the figure.



(A) The magnetic field at 'O' due to current carrying loop is $\frac{\mu_0 I}{2\pi R}\ell n(\sqrt{2}+1)(\hat{k})$
(B) The magnetic field at 'O' due to current carrying loop is $\frac{\mu_0 I}{\pi R}\ell n(\sqrt{2}+1)(\hat{k})$
(C) The magnetic moment of the system is $4R^2 I(2+\pi)(+\hat{k})$
(D) The magnetic moment of the system is $2R^2 I(2+\pi)(+\hat{k})$